Data Storytelling

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Spring 2026

Week 05

Announcements

- Problem Set 1 is due next week (Week 06) on **Monday March 2nd at 11:59 PM**.
  - How's it coming along? Any questions?

Survey Data

The Measurement Problem

- We often want to measure concepts that are not directly observable, such as:
  - Public opinion (e.g., “How would you rate the current state of the economy?”)
  - Political attitudes (e.g., “Do you approve of the job that the president is doing?”)
  - Vote intentions (e.g., “Who will you vote for in the next election?”)
- However, outside of the census, we rarely have access to data for every individual relevant to our study.
  - How can we get accurate measures of a population without access to population-scale data?

Samples

- Typically, in social science research we deal with `samples` rather than populations.
  - A *sample* is a subset of individuals from a population that is used to make inferences about the population.
  - A *population* is the entire group of individuals that we are interested in studying.
  - We use samples to measure *parameters* (μ) of the population, which we call *statistics*($\bar{x}$).

Sampling: Parameters vs Statistics

Illustrative Example of Parameters and Statistics 1

- Let’s say we want to know the average height of all adults in the United States. We can’t measure the height of every adult, so we take a sample of 1,000 adults and measure their heights.
  - If the average height of an adult man is 5’9” nationwide, that is our ______ of interest.
  - If the average height of our *sample* is 5’8.3”, that is our estimated ______.
- Why would they not be the same?

Illustrative Example of Parameters and Statistics 2

- Let’s say we don’t have the resources to take a national sample and instead sample of 500 Olympic athletes and measure their heights.
  - Their average height is 6’2”
  - Can we use this sample to estimate the average height of all adults in the United States? Why or why not?

Random Error vs Systematic Error

- **Random error**
  - A natural byproduct of the sampling process.
  - The difference between the sample statistic and the population parameter that occurs by chance.
  - Random error can be reduced by increasing the sample size.
- **Systematic error**
  - Bias that occurs when the sample is not representative of the population.
  - Systematic error can be caused by factors such as selection bias, non-response bias, or measurement error. Systematic error cannot be reduced by increasing the sample size and can lead to inaccurate parameter estimates.

Law of Large Numbers

- The Law of Large Numbers states that as the sample size increases, the sample statistic will converge to the population parameter.

$$\lim_{n \to \infty} \bar{x} = \mu$$

  - In other words, as we take larger and larger samples, the average of the sample will get closer and closer to the true average of the population.

Law of Large Numbers

- This is why random sampling is so powerful:
  - Even if we can't measure the entire population, we can still get accurate estimates of population parameters by taking a sufficiently large random sample.
  - This effect is non-linear: random error decreases rapidly as sample size increases
- *Note:* the Law of Large Numbers does *not* converge to the true parameter if there is systematic error in the sampling process.

Correlational Analysis

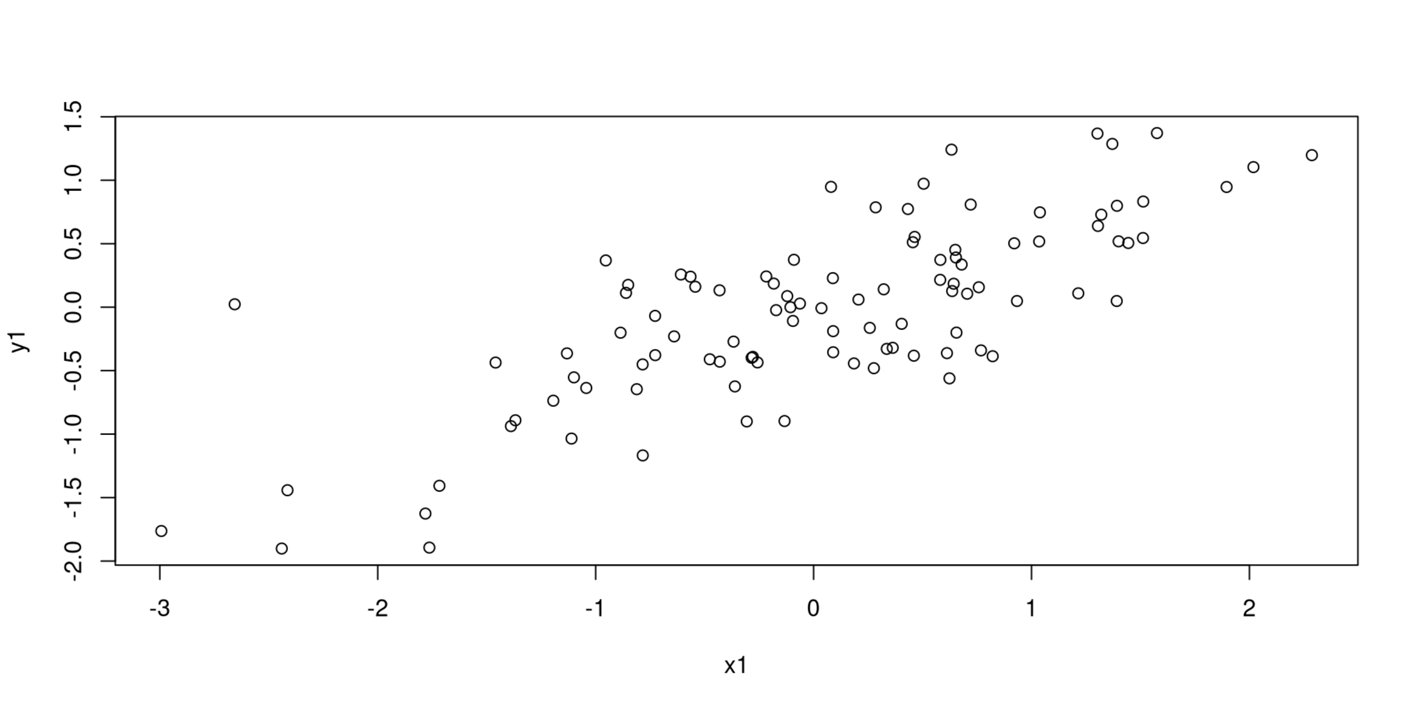
Correlation (ρ)

- Given a relationship of interest, we oftentimes want to know to what extent they are related to one another and how strong that relationship is.
  - For example, we might want to know the relationship between education and income, or the relationship between political ideology and voting behavior.
- Correlation is a statistical measure that describes the strength and direction of a relationship between two variables.
  - Correlation is denoted by the Greek letter ρ (rho) for population correlation and by the letter r for sample correlation.

Correlation Coefficient

- The correlation coefficient (ρ) is a measure of the strength and direction of a linear relationship between two variables.
  - r ranges from -1 to 1
  - $r = 0$ indicates no correlation
  - $r > 0$ indicates a positive correlation
  - $r < 0$ indicates a negative correlation

A few examples of correlation coefficients:



- What do you think the correlation coefficient is for this relationship?
  - .76

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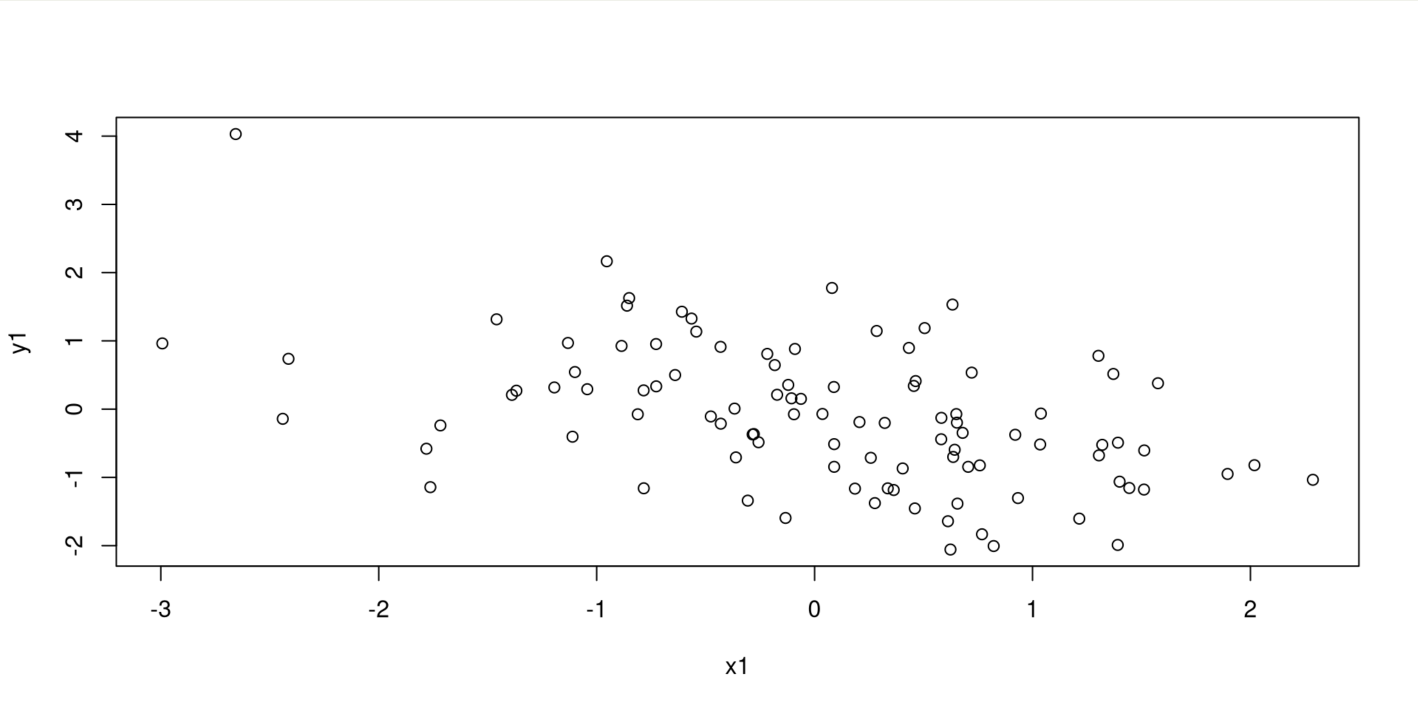
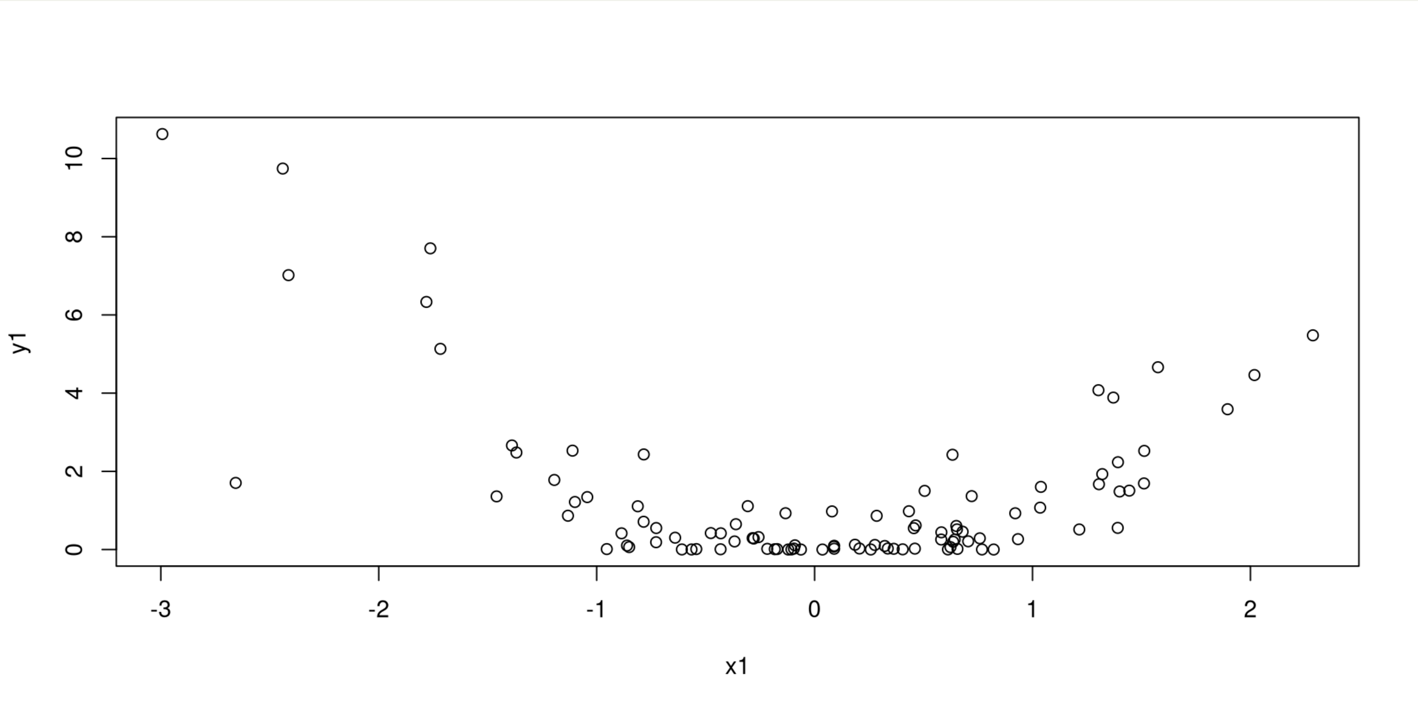


- What do you think the correlation coefficient is for this relationship?
  - -.47

A few examples of correlation coefficients:



- What do you think the correlation coefficient is for this relationship?
  - -.29

Exercise: Analyzing Survey Data

(see code)